

STUDENT ATTENDANCE TRACKING SYSTEM

Complete App Creation and Feature Guide

Main Management App + Parent Mobile App

Prepared for HamdhyTech

Final low-cost design - no student photo storage - optional SMS only

Document Purpose

This guide defines the complete functional and technical plan for a two-application student attendance system. The Main App is used by the HamdhyTech Super Admin, Institute Admins and Teachers. The Parent App is used by parents or guardians to view attendance and receive updates.

Confirmed cost direction

The system is designed to avoid paid student photo storage and unnecessary services. Student photos are excluded. Firebase free-tier services, free development tools and direct APK installation should be used wherever possible. Optional SMS through a provider such as Notify.lk is the only planned recurring paid feature. Free-tier quotas, provider policy changes, internet/data charges and app-store publishing fees remain outside this guarantee.

Final System Structure

- HamdhyTech Super Admin creates and controls institutes.
- Each Institute Admin creates teachers, classes and students.
- Teachers use continuous QR scanning to record arrival and departure.
- Parents link their child using institute code, student number and verification.
- Attendance updates appear inside the Parent App in real time.
- Push notifications are used where the chosen free backend setup supports them; SMS is optional and controlled by the Super Admin.
- Parent mobile numbers are mandatory for student registration and SMS/verification purposes.

Role Hierarchy

Role	Created By	Main Access	Data Scope
HamdhyTech Super Admin	System owner	All institutes, permissions, SMS controls, system monitoring	Whole system
Institute Admin	Super Admin	Teachers, classes, students, attendance reports	Own institute
Teacher	Institute Admin	Assigned classes, QR attendance, student management if allowed	Assigned institute/classes
Parent/Guardian	Self-registration or institute invitation	Linked children, attendance and alerts	Linked students only

SECTION 1 - MAIN MANAGEMENT APP

The Main App is one mobile application with role-based dashboards. The same login screen can authenticate the Super Admin, Institute Admin and Teacher, but the backend role determines which dashboard and permissions are shown.

1. Login and Account Security

- One clean login screen with username or email and password.
- Backend role validation after login; never trust a role selected only on the screen.
- Super Admin credentials must be stored securely through authentication and must never be hardcoded in the APK.
- Institute Admin and Teacher accounts use temporary passwords and must change them at first login.
- Account statuses: Active, Suspended, Locked and Password Reset Required.
- Password reset must replace the password; old passwords must never be displayed.

Security correction

A username such as HamdhyTechAdmin can be used, but the actual production password should be stronger and different from any password shared in conversation or documentation.

2. HamdhyTech Super Admin Dashboard

The Super Admin is the system owner. No public user can create this role. The dashboard gives complete control of institutes and service permissions.

- Create, edit, activate, suspend or archive institutes.
- Create the first Institute Admin account and issue a temporary password.
- View institute name, code, status, teacher count, class count, student count and daily attendance activity.
- Open any institute to view its teachers, classes, students and attendance summary.
- Enable or disable Parent App access, push notifications and SMS independently.
- Set SMS monthly allowance, current usage, extra-SMS permission and warning thresholds.
- View SMS delivery reports, failed notifications and system audit logs.
- Reset Institute Admin credentials and force password changes.
- Configure package status or expiry date when the system is commercially offered.

3. Institute Creation

The Super Admin creates an institute before any teacher can use the system. Recommended fields are listed below.

Field	Purpose
Institute name	Displayed in the app, reports, push notifications and SMS messages.
Institute code	Unique code used by parents and internal records.
Address and contact number	Shown to parents and used for support.
Institute Admin name	Responsible manager for the institute.
Admin username/email	Used for secure login.
Temporary password	Changed at first login.
SMS enabled	Controls whether the institute may send paid SMS.
Push enabled	Controls app notification functionality.

Account status/expiry	Controls system access.
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4. Institute Admin Portal

The Institute Admin manages only their own institute. This role organizes staff, classes, students and reports but cannot change system-wide permissions controlled by HamdhyTech.

- Edit institute profile and contact information.
- Create, edit, disable and reset Teacher accounts.
- Create classes, subjects, grades, schedules and room details.
- Assign one or more teachers to classes.
- Add students manually or import a prepared CSV file.
- Assign students to one or more classes.
- View all attendance sessions and make approved corrections.
- Approve or reject parent-student linking requests.
- View SMS status but not override a Super Admin restriction.
- Export attendance reports as CSV or PDF.

5. Teacher Account and Permissions

The Institute Admin creates each Teacher account. Teacher permissions can be adjusted so that different institutes can choose stricter or more flexible workflows.

- View assigned classes and today's class schedule.
- Create classes only when permission is enabled.
- Add or edit students only when permission is enabled.
- Generate and share student QR cards.
- Open continuous QR attendance mode.
- Mark entry, departure, late, absent or excused status.
- View attendance history for assigned classes.
- Correct a record only with a required reason.
- Export class-level reports when allowed.

6. Student Registration - No Photo Storage

Student photos are deliberately excluded to avoid cloud image storage requirements and additional costs. Identity is shown using the student name, student number, class and QR status.

Mandatory student information

- Full student name.
- Student number entered by the institute; automatically generated only when left blank.
- Institute ID and institute code.
- Assigned class or classes.
- Primary parent/guardian name.
- Primary parent/guardian mobile number - mandatory.
- Secondary parent/guardian name and mobile number - optional.
- Preferred notification language: English, Sinhala or Tamil.
- Push notification enabled/disabled.
- SMS enabled/disabled, subject to institute permission and parent consent.
- Student status: Active, Inactive, Graduated or Suspended.

Mobile number requirement

The parent mobile number must be collected and validated. It supports parent identification, student linking, optional SMS, emergency contact and account recovery. The number must be visible only to authorized staff.

7. Class Management

- Class name, class code, subject, grade and academic year.
- Assigned teacher or teachers.
- Days, start time, expected end time and location.
- Active student list and total enrolment.
- Attendance settings such as late threshold and whether departure scanning is required.
- Class status: Active, Completed, Cancelled or Archived.

8. Student QR Code

Each student receives a unique QR code. The QR must contain a secure token or reference, not the parent mobile number or other private details. The app uses the token to retrieve the student record securely.

- Display QR inside the Main App.
- Generate a printable or shareable PDF QR card.
- Include institute name, student name, student number and QR code on the card.
- Regenerate a QR code if it is lost or copied.
- Disable the previous QR token immediately after regeneration.
- Prevent students from another institute or unassigned class from being accepted.

9. Continuous Automatic QR Attendance

The teacher must not press a scan button for every student. The scanner opens once and stays active until the teacher ends the attendance session.

1. Teacher logs in and selects the class.

2. Teacher chooses Entry Mode or Departure Mode.
3. Teacher taps Open Scanner once.
4. The camera continuously detects QR codes.
5. When a valid student QR is detected, attendance is saved automatically without a confirmation tap.
6. The latest student details appear in a panel below the camera.
7. A success sound, vibration and green indicator confirm the record.
8. After a short delay, the panel clears while the camera remains ready for the next student.
9. The teacher ends the session only after all students are processed.

Scanner screen information

- Class name and current mode at the top.
- Live camera area in the centre.
- Present/scanned count, for example 24 of 32.
- Latest student name, number, recorded time and status below the camera.
- Large green success state for accepted QR codes.
- Red warning for invalid, inactive, wrong-class or duplicate QR codes.
- Pause, flash, switch camera and End Session controls.

Example success panel

Attendance recorded

Student: Mohamed Farhan

Student No: ST-1045

Class: Grade 10 Mathematics

Status: Arrived

Time: 3:57 PM

10. Attendance Validation Rules

- Duplicate scans within a configurable period are ignored and reported as Already Recorded.
- Departure cannot be recorded before arrival unless an Institute Admin corrects the record.
- A student not assigned to the selected class is rejected.
- Inactive or replaced QR tokens are rejected.
- Each record stores class, institute, date, event type, time, teacher and device reference.
- Manual changes require a reason and are added to the audit log.
- Offline records may be queued locally and synchronized when internet access returns, but SMS or push alerts will wait until synchronization.

11. Attendance Statuses

Status	Meaning
Arrived	Entry QR successfully recorded.
Departed	Exit QR successfully recorded.
Late	Arrival occurred after the configured late threshold.
Absent	No valid arrival was recorded for the session.
Excused	Absence accepted by authorized staff.
Incomplete	Arrival recorded but departure missing when departure is required.

12. Notifications and SMS Controls

Attendance recording and message delivery are separate. Attendance must always be saved even when SMS or push notifications are disabled.

Decision flow

10. Save the attendance record.
11. Update the Parent App attendance status.
12. Check whether push notifications are enabled for the institute and parent.
13. If enabled and technically available, send an app push notification.
14. Check whether SMS is enabled by the Super Admin, enabled for the student and accepted by the parent.
15. If all SMS conditions are true and allowance is available, send the optional paid SMS.
16. Save notification and SMS delivery results in the log.

Attractive message format

Every message should begin with the institute name, followed by a clear and friendly attendance update. Messages should be short to reduce the risk of being counted as multiple SMS units.

Arrival message

BRIGHT FUTURE ACADEMY

Dear Parent, Mohamed Farhan arrived for Grade 10 Mathematics at 3:57 PM. Thank you.

Departure message

BRIGHT FUTURE ACADEMY

Dear Parent, Mohamed Farhan departed from Grade 10 Mathematics at 6:04 PM. Thank you.

Late arrival message

BRIGHT FUTURE ACADEMY

Dear Parent, Mohamed Farhan arrived late for Grade 10 Mathematics at 4:18 PM.

- Institute name is automatically inserted.
- Student name, class name, date and time are inserted dynamically.
- Institute Admin may customize wording within safe templates.
- English, Sinhala and Tamil templates may be supported.
- Long Unicode SMS can consume more than one SMS unit; preview the estimated unit count before saving a template.

13. SMS Administration

- Super Admin switch: Allow SMS for this institute.
- Institute-level monthly SMS allowance and usage counter.
- Warning at 70%, 80%, 90% and 100% usage.
- Allow paid extra SMS: ON/OFF.
- Arrival SMS and departure SMS can be enabled separately.

- Student-level SMS permission and parent consent.
- Delivery status: Sent, Delivered, Pending, Failed or Skipped.
- When SMS is disabled, Teacher App displays “SMS is not enabled for this institute; attendance will still be recorded.”

14. Reports and Exports

- Daily attendance report by class.
- Monthly attendance percentage by student.
- Late arrivals and early departures.
- Missing departure records.
- Attendance corrections and audit report.
- SMS usage and delivery report.
- CSV export for opening in Google Sheets.
- PDF report for printing or sharing.
- Google Sheets should be optional reporting only, not the primary database.

15. Main App User Interface Principles

- Large buttons and simple wording for teachers during busy attendance sessions.
- Use consistent icons for Institutes, Teachers, Classes, Students, Scanner, Reports and Settings.
- Keep the scanner screen free of unnecessary menus.
- Use clear green success, amber warning and red failure states with text labels.
- Dashboard cards should show today’s classes, total students and attendance totals.
- Confirm destructive actions such as deleting a class or regenerating a QR code.
- Do not expose parent mobile numbers on public screens or QR cards.

SECTION 2 - PARENT MOBILE APP

The Parent App must be especially simple and friendly. A parent should be able to open the app and immediately understand whether the child has arrived, is currently in class or has departed.

16. Parent Registration and Login

- Register using parent name, mobile number, email if available and password.
- Use email/password or username/password to avoid paid phone-OTP charges in the low-cost version.
- Request notification permission during onboarding with a clear explanation.
- Allow one parent account to link multiple children.
- Allow a child to be linked to approved primary and secondary guardians.
- Provide account recovery through institute-assisted reset or email when configured.

17. Secure Student Linking

Institute code and student number alone are not sufficient because another person may know them. Add a secure linking verification method.

17. Parent enters Institute Code.
18. Parent enters Student Number.
19. Parent enters a linking PIN issued by the institute, or submits a linking request.
20. The system compares the parent mobile number with the number registered for the student.

21. Institute Admin approves the request when required.
22. After approval, the child appears on the Parent App dashboard.

Recommended verification

Use the registered parent mobile number plus a teacher-issued linking PIN. This avoids paid OTP SMS while still preventing casual unauthorized access.

18. Parent Home Dashboard

The dashboard should display the most important information without requiring the parent to open several menus.

- Student name and student number - no photo required.
- Current status: Not Arrived, In Class, Departed, Late or No Class Today.
- Current or next class name and expected time.
- Today's arrival time and departure time.
- Latest attendance update in a prominent card.
- Today's class list.
- Quick link to attendance history and institute contact details.
- Child selector at the top when multiple children are linked.

19. Real-Time Attendance Updates

When the teacher scans a QR code, the attendance record is updated in the central database. The Parent App listens to the linked student's records and refreshes the screen automatically while the app is open.

- Arrival time appears immediately after synchronization.
- Status changes from Not Arrived to In Class.
- Departure changes the status to Departed.
- When push delivery is available, the parent also receives a phone notification similar to a messaging-app alert.
- When SMS is enabled, a separate SMS is sent through the configured paid SMS provider.
- If internet is unavailable, the app shows the most recently synchronized information and the time it was last updated.

20. Parent Notifications Centre

- Arrival notification.
- Departure notification.
- Late arrival notification.
- Class cancellation or schedule change, when added later.
- Unread notification badge.
- Notification history grouped by date and child.
- Tapping an alert opens the correct student and attendance record.
- Parent settings for arrival, departure and late alerts.

21. Attendance History

- Daily timeline with class, arrival and departure.

- Calendar view showing present, absent, late and excused days.
- Monthly attendance percentage.
- Filter by child, class and date range.
- Clear explanation for corrected records.
- Option to download or share a simple attendance summary later.

22. Parent App User-Friendly Design

- Use a simple bottom navigation: Home, History, Notifications and Settings.
- Large readable text and clear status labels.
- Use both color and words so the app remains understandable for users with color-vision difficulties.
- Keep technical words such as token, sync and database out of the parent interface.
- Show institute contact number clearly for help.
- Allow English, Sinhala and Tamil interface text in future versions.
- Minimize typing during student linking.
- Provide clear messages when SMS is unavailable but app updates are still active.

SECTION 3 - SHARED TECHNICAL DESIGN

23. Recommended Technology

- Flutter for both Android mobile applications using shared models and services.
- Firebase Authentication for secure account login.
- Cloud Firestore or Firebase Realtime Database for structured attendance data within free-tier limits.
- Firebase Cloud Messaging for app push notifications where the selected backend arrangement supports secure sending.
- Local encrypted storage for session information and offline queues.
- QR generation, QR scanning, PDF generation and CSV export using free Flutter packages.
- Notify.lk or another SMS API only when the institute enables the paid SMS feature.

24. Suggested Database Collections

Collection	Main purpose
institutes	Institute profile, code, status, permissions and SMS settings.
users	Super Admin, Institute Admin, Teacher and Parent role records.
classes	Class details, schedule and assigned teachers.
students	Student name, number, parent contacts and notification preferences. No photo field required.
classStudents	Student-to-class assignments.
attendanceSessions	Each class attendance session.
attendanceRecords	Arrival, departure, time, status and audit information.
parentStudentLinks	Approved guardian relationships.
devices	Parent device notification tokens.
notifications	In-app and push message history.
smsLogs	SMS request, delivery result and SMS unit usage.
auditLogs	Account, student and attendance changes.

25. Important Data Fields

- Every record must contain `instituteId` to separate institute data.
- Attendance records contain `classId`, `studentId`, `sessionId`, event type, timestamp, `teacherId` and sync status.
- Parent contact data must be protected by role-based security rules.
- QR tokens must be revocable and must not expose parent numbers.
- SMS settings should exist at system, institute and student/parent levels.
- Timestamps should be generated by the trusted server where possible, not only by the teacher's phone clock.

26. Security Requirements

- Role-based access enforced by backend security rules.
- Institute data isolation: users cannot read another institute's students or attendance.
- Parents can read only approved linked-student data.
- Teachers can access only assigned or permitted classes.
- Passwords are handled by authentication, never stored as plain text.
- SMS provider API keys and notification server credentials must not be placed in the mobile APK.
- Keep an audit trail for account creation, QR regeneration and attendance correction.
- Use minimum necessary personal data and provide a process to deactivate or delete records.

27. Low-Cost Plan, Free Quotas and Limits

The system is designed so that optional SMS is the only planned recurring service charge. This is achievable for a small deployment when usage remains inside free quotas, student photos are not stored, phone-number OTP is avoided and the apps are distributed directly as APK files. The following limits must still be monitored.

Component	Free availability	Free quota or limit	What happens at the limit
Flutter, Dart, Android Studio and VS Code	Free development tools	No normal usage quota	No service charge. Development still requires a computer and internet connection.
Email/password authentication	Available on the Firebase no-cost plan	Suitable for normal username/email and password sign-in. Do not use paid phone OTP for this design.	Abusive or unusually high usage may be restricted under provider policies.
Cloud Firestore database	One free database per Firebase project	1 GiB stored data; 50,000 document reads/day; 20,000 writes/day; 20,000 deletes/day; 10 GiB outbound transfer/month. Daily operation quotas reset around midnight Pacific time.	On the no-cost plan, operations that need more quota will not continue until quota resets or billing is enabled. Optimize live listeners and reports to reduce reads.
Firebase Cloud Messaging push notifications	FCM itself is no-cost	Default downstream quota is about 600,000 messages per minute per project, far above a normal attendance app. A single Android device is limited to 240 messages/minute and 5,000/hour.	Over-quota requests can return an error and must be retried. These limits are subject to change.
Secure notification sender	Can be kept free only when hosted within a suitable free allowance	A trusted backend is required to securely call FCM. Never place server credentials in the mobile app.	Some Firebase server products require the Blaze billing plan. Use a carefully selected free-tier backend or accept that billing activation may be required even when actual use remains free.
Student photos / Firebase Storage	Not used in this project	No photo storage, upload or download quota is required.	Cloud Storage for Firebase requires the Blaze plan as of February 3, 2026; therefore photos are deliberately excluded.
QR code generation and scanning	Free Flutter packages	No provider message quota; processing occurs on the phone	Device performance and camera quality are the practical limits.
PDF QR cards and CSV reports	Generated locally using free packages	No cloud quota when files are created on the device	Large reports use device storage and memory.
Real-time Parent App updates	Uses Firestore listeners	Included inside the Firestore read and transfer quotas above	Leaving many broad listeners active increases reads. Listen only to the linked student and recent attendance records.
Direct APK distribution	Free	Unlimited direct sharing/installing where Android security settings permit	Updates must be distributed manually and users must trust the installation source.
Google Play publishing	Optional, not required	No free developer registration	Google currently charges a one-time US\$25 Play Console registration fee.
Notify.lk SMS	Optional paid feature	Lite package described by the user includes 450 SMS/month; extra SMS is charged by the provider	Monthly package fees, extra-message charges, taxes and Unicode multi-part SMS rules may apply.
Mobile internet/data	Not supplied by the app	Depends on each teacher/parent mobile plan or Wi-Fi	The user or institute pays their own internet/data cost.

How the Free Database Quota Relates to Attendance

One attendance scan normally creates or updates only a small number of documents. For example, 500 students generating an arrival and departure event on one day would create about 1,000 core attendance writes, before audit or notification records. This is below the 20,000-write daily free quota. The larger risk is unnecessary reads from dashboards and real-time listeners, so the Parent App must query only linked students and recent records instead of downloading complete institute data.

Cost-Control Rules for This Project

- Keep student photos completely disabled so Firebase Storage is not required.
- Use email/username and password login; do not use SMS OTP authentication.
- Use FCM push notifications as the default parent alert because FCM itself is free.
- Enable Notify.lk SMS only for institutes that have an active paid SMS package.
- Keep the SMS switch separate for system, institute and student/parent consent levels.
- Generate QR PDFs and reports on the device instead of uploading them to cloud storage.
- Limit Firestore listeners to the current class or linked student and close listeners when screens are not visible.
- Use pagination and date filters for attendance history instead of loading all records.
- Show database usage and SMS usage warnings in the Super Admin dashboard.
- Distribute APK files directly during testing to avoid Play Store registration costs.

Accurate cost statement

For the planned small version, SMS can be the only intentional recurring service payment when the application remains within free quotas and a suitable no-cost notification sender is available. However, internet/data costs, optional Play Store registration, provider policy changes and usage above free quotas are external costs that cannot be guaranteed as permanently free.

Pricing and quota information checked on 1 August 2026 using official Firebase and Google Play documentation. Providers may change limits, requirements and prices, so recheck them before production launch.

28. Development Phases

23. Create Flutter workspace, shared package and Firebase project.
24. Build authentication and role-based routing.
25. Build Super Admin institute management.
26. Build Institute Admin teacher, class and student management.
27. Build QR creation and PDF sharing.
28. Build continuous automatic QR scanner and validation rules.
29. Build attendance history, reports and CSV export.
30. Build Parent registration and secure student linking.
31. Build Parent dashboard and real-time attendance updates.
32. Add push notifications using a secure sender architecture.
33. Add optional Notify.lk SMS and Super Admin controls.
34. Complete security rules, testing, backups and deployment.

29. Testing Checklist

- Super Admin cannot be self-created by a normal user.
- Institute Admin cannot access another institute.
- Teacher cannot access an unassigned class.
- Parent cannot link a student without correct verification.
- Continuous scanner records multiple students without reopening the camera.
- Duplicate QR scans do not create duplicate attendance.

- Wrong-class and inactive QR codes are rejected.
- Attendance saves even when SMS is disabled.
- Parent App updates after attendance synchronization.
- SMS is sent only when all permission levels are enabled.
- No student photo upload or cloud photo field exists.
- Parent mobile numbers are protected from unauthorized users.
- Offline records synchronize safely without duplicates.
- Reports match the original attendance records and corrections.

30. Final Minimum Viable Product

Main App MVP

- Secure role login.
- Super Admin institute creation and SMS control.
- Institute Admin teacher, class and student management.
- Mandatory parent mobile number collection.
- Student QR generation and PDF sharing.
- Continuous automatic QR entry/departure scanning.
- Daily attendance history and corrections.
- CSV/PDF reports.
- Optional SMS integration.

Parent App MVP

- Parent account registration.
- Secure student linking.
- Friendly home dashboard.
- Real-time arrival and departure status.
- Attendance history.
- In-app notification history.
- Push alerts where configured.
- Multiple-child support.

Final Approved Workflow

Complete flow

HamdhyTech Super Admin creates an institute -> Institute Admin creates teachers, classes and students -> Parent mobile number is recorded -> Student QR card is generated -> Teacher opens continuous scanner once -> Each student is recorded automatically -> Attendance updates the Parent App -> Push notification is sent where configured -> Optional SMS is sent only when permitted and paid SMS service is enabled.

Conclusion

This final design produces a practical, scalable and parent-friendly attendance system without student photo storage. The Main App provides controlled administration and fast automatic QR attendance, while the Parent App provides immediate, simple attendance visibility. The architecture keeps SMS optional,

protects parent information, separates institute data and allows the project to begin with free development tools and free-tier services.